

Aditi Ganapathi

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Education

Arizona State University

May 2025 – Present

PhD in Computer Science

Research Areas: AI Privacy, Usable Security, Adversarial Machine Learning, Human-ML Interaction

Arizona State University

Aug 2023 – May 2025

MS in Computer Science | GPA: 3.81/4.0

Coursework: NLP, Data Mining, High-performance Computing, Human-ML Interaction, Software Security

Amity University

Jul 2019 – Jun 2023

B.Tech in Computer Science & Engineering | CGPA: 9.01/10.0

Coursework: Artificial Intelligence, Deep Learning, Discrete Math, Probability & Statistics, Data Structures

Experience

HAPPY Lab, Arizona State University

Jan 2025 – Present

Research Assistant

Tempe, AZ, USA

- Conducted human-centered cybersecurity and privacy research mentored by **Dr. Jaron Mink** in collaboration with **SEFCOM** Lab.
- Designed a unified framework with **fine-tuned LLMs** to detect vulnerabilities in LLMs for **adversarial prompting** across both proprietary and open-weight models (**GPT-4o**, **LLaMA**, **Grok**, **Gemini**, **Mistral**, etc).
- Applied gradient-based adversarial techniques like **Fast Gradient Search Method (FGSM)** and **Projected Gradient Descent (PGD)** to generate visual adversarial examples that embed hidden prompts and meta-instructions into images.

Amazon Web Services

Jan 2023 – Jun 2023

Software Engineer Intern

Bengaluru, KA, India

- Engineered and deployed a new security model for **AWS Aurora's** Canary Management Service, integrating advanced **IAM**, **Bindles**, **POSIX**, and **LDAP** mechanisms for enhanced authentication and authorization.
- Integrated **Cognito** and **Midway** with **API Gateway** and **Lambda** for proper authentication while ensuring seamless service request handling.
- Utilized **pico-cli** to streamlining operational workflows and system interactions, **increasing user engagement by 30%**.
- Expanded service availability to **two new partitions**, impacting **over 1,000** existing customers.

Selected Projects

Compositional T2I Evaluation with Description Logic and LoRA

- Investigated limitations in **DALL-Eval** and **HRS-Benchmark** to identify biases in prompt generation pipelines and failure cases in compositional reasoning.
- Developed a **Description Logic (DL)-based evaluation** framework to systematically generate compositional prompts.
- Designed an automated evaluation pipeline using **LoRA fine-tuned Stable Diffusion v1.4** and **v2.1**, evaluating attribute consistency, object interaction correctness, and hallucination rates in **reduced time and memory**.

VectorDB-based Retrieval Augmentation

- Integrated Vector Databases (**FAISS** and **ChromaDB**) with **RAG** models using **Zephyr Beta 7B**.
- Compared vector search methods including **IVFL2**, **IVFFLAT**, and **HNSW** in FAISS, and ChromaDB's **hybrid search** capabilities to evaluate retrieval accuracy, speed, and computational efficiency.
- Applied indexing strategies like **Voronoi cell-based partitioning** and **dense-sparse retrieval** techniques.

Skills

Languages:	Python, C++, C, Java, JavaScript, SQL, MATLAB, Shell, Linux, Git
ML Methods:	Language Models, RAG, Multi-Modal Learning, Fine-Tuning, Transfer Learning, Adversarial Machine Learning, Vector Search & Indexing
ML Frameworks:	PyTorch, Lightning, TensorFlow, HuggingFace, LangChain, Gradio, Stable Diffusion, scikit-learn, FAISS
Python Libraries:	NumPy, Pandas, SciPy, spaCy, NLTK, Transformers, Gensim, OpenCV, Matplotlib, Seaborn, Pillow